PROJECT ENGINEERING REPORT

Continuum

Quality fixes and verification

200 additional corrections

Timed assessments and collaborative learning.

Classroom management, teacher authoring, and application recovery.

23 October 2026

Baseline revision

`1ec26fb0b59cb4704ccce28b93fc4046da5a35f4`

Technical supplement to the project and quality audit.

1 Project context

Continuum uses a Go backend and a React/TypeScript frontend. The backend separates HTTP transport, authentication, assessment rules, classroom operations, collaborative boards, storage, health checks, and recovery coordination. DynamoDB and in-memory stores provide the persistence interfaces. The frontend uses TanStack Router and Query, Base UI components, and Excalidraw for collaborative canvases.

The correction batch concentrates on browser state, asynchronous operations, validation, telemetry, and interaction quality. The deployment configuration and live AWS resources are outside this change.

2 Scope and outcomes

The register contains 200 additional correction entries covering behavioral defects, invalid-data contracts, recovery edge cases, accessibility, and polish. Related entries can share a code path while having different triggers or visible outcomes. The total does not represent 200 production incidents or 200 independent root causes.

The earlier 120-item audit remains a historical baseline. Its findings are not all marked resolved by this batch. The fixed-calendar authentication tests identified as CQ-119 were also repaired as a verification prerequisite; that repair is excluded from the 200-entry count.

Area	Corrections
Identity and API	41
Navigation, assessment UX and access	49
Authoring	33
Boards and history	21
Recovery and telemetry	25
Classrooms	31
Total	200

Material behavior changes

Browser restoration checks the shape and identity of saved records before using them. Pending submissions preserve answer snapshots across retries, and recovery selects the latest attempt. Draft identities remain distinct when saves happen in the same millisecond.

Classroom operations reject account changes between preflight and mutation. Sequence checks prevent older list and roster responses from overwriting newer state. Details updates preserve concurrently changed membership.

Quiz settings retain unsaved edits during catalog refreshes. Pending dialogs prevent overlapping actions and accidental dismissal. Teacher previews show complete options, weights, feedback, and deliberate line breaks.

HTTP and recovery checks reject malformed measurements and responses without producing invalid JSON. Routing applies a deadline to control-state reads, and proxies and health checks stop following redirects automatically. SMTP messages use valid envelopes and MIME encoding for international text.

3 Verification

The full Go package suite passes with the race detector and a 30-minute test timeout. Go vet passes. The frontend passes type checking, lint, the production build, 238 unit/component tests across 30 files, and eight Playwright journeys.

New regression coverage exercises corrupted storage, identity changes during requests, response ordering, retry payload stability, imported data, JWT boundaries, request parsing, telemetry values, HTTP status capture, redirect handling, Unicode email delivery, previews, dialogs, and navigation inputs.

The automated journeys run against isolated local services. These checks do not establish production deployment behavior, live email-provider interoperability, or exhaustive assistive-technology coverage. Existing production-build warnings about large chunks and route exports remain.

4 Correction register

The numbered entries correspond to the machine-readable JSON and Markdown registers in the repository. File headers identify the relevant implementation. Source and regression tests are included in the commit.

ID	Corrected behavior
`web/src/lib/auth.ts`	
QF-001	Blank persisted user IDs created an authenticated session
QF-002	Unknown persisted roles were silently changed to student
QF-003	Non-text profile fields were coerced into misleading display strings
QF-004	Partial token records produced authenticated pages without a usable session
QF-005	Caller mutations silently changed the auth snapshot
QF-006	Unrelated session-storage events reloaded local auth
`web/src/lib/api.ts`	
QF-007	Trailing slashes and whitespace in API origins produced malformed URLs
QF-008	Headers objects and tuple headers lost their contents
QF-009	Lowercase authorization headers were overwritten
QF-010	Public authentication requests carried the previous user's bearer token
QF-011	A rejected login retried through the old account's refresh token
QF-012	Malformed refresh responses replaced valid credentials
QF-013	Top-level API error codes disappeared
QF-014	Error-string responses lost their actionable message
QF-015	Successful 205 responses failed JSON parsing
`web/src/lib/pending-submission.ts`	
QF-016	Array-shaped pending answers passed object validation
QF-017	Noncanonical saved question keys changed answer identity on reload
QF-018	Fractional or negative saved choices survived restoration
QF-019	Invalid pending attempt numbers escaped scope checks
QF-020	Blank generated request IDs created unusable retry records
QF-021	Invalid pending timestamps corrupted recovery ordering
QF-022	Pending records stored under a different attempt key were recovered
QF-023	An older attempt with a later timestamp hid a newer pending attempt
QF-024	Invalid retry counts ended without the expected final error
QF-025	Invalid retry delays overflowed browser timers
QF-026	A submit callback could mutate the next retry's payload
QF-027	HTTP 408 submission failures were not retried
QF-028	HTTP 500 submission failures were not retried
`web/src/lib/quiz-drafts.ts`	
QF-029	Malformed nested draft questions crashed the builder
QF-030	Malformed optional draft settings reached rendering and API requests
QF-031	Duplicate stored draft IDs rendered duplicate controls
QF-032	Drafts saved in the same millisecond overwrote each other

ID	Corrected behavior (continued)
`web/src/lib/question-bank.ts`	
QF-033	Blank bank identities made items impossible to target reliably
QF-034	Invalid bank timestamps rendered invalid dates
QF-035	Malformed option lists reached the question builder
QF-036	Out-of-range saved correct answers could be inserted
QF-037	Malformed saved scoring or feedback fields broke authoring
`web/src/lib/board-history.ts`	
QF-038	Blank room identities appeared in board history
QF-039	Invalid history timestamps broke sorting and date display
QF-040	Invalid visit counts rendered nonsensical history
QF-041	Duplicate saved board rows produced duplicate controls
QF-042	History was truncated before selecting the most recent visits
QF-043	Board history stayed stale when another tab changed it
QF-044	Visit counters could overflow safe integer precision
QF-045	History merge sorted timezone strings instead of instants
QF-046	Equal-time board rows had unstable ordering
QF-047	Downloads clicked a detached anchor
QF-048	Downloads revoked their blob before the browser consumed it
`web/src/lib/recent-rooms.ts`	
QF-049	Empty recent-room entries created broken navigation
QF-050	Duplicate recent rooms displaced usable history
QF-051	Recent-room changes in another tab were not reflected
`web/src/lib/quiz-progress.ts`	
QF-052	Invalid question counts produced invalid restored positions or unbounded arrays
QF-053	Invalid progress records were reported as successfully persisted
QF-054	Invalid flag targets polluted question navigation
`web/src/lib/assessment-history.ts`	
QF-055	Unknown attempt statuses bypassed coherent status filtering
QF-056	Invalid assessment dates destabilized sorting
QF-057	Invalid retake counters enabled impossible retakes
`internal/auth/auth.go`	
QF-058	Email local parts with repeated or boundary dots were accepted
QF-059	Invalid DNS labels were accepted as email domains
QF-060	Overlength emails bypassed validation outside HTTP registration
QF-061	The auth service itself allowed invalid email identities
QF-062	Missing profile updates returned a successful null user
QF-063	Token issuance panicked on a missing user
QF-064	Unknown token types could be minted
QF-065	Subsecond token lifetimes minted already expired credentials
QF-066	Signed JWTs without expiration were accepted
QF-067	JWTs issued in the future were accepted
QF-068	Nil decoded tokens panicked
QF-069	Whitespace-only JWT subjects passed subject validation
QF-070	Fractional, negative or overflowing session versions were accepted

ID	Corrected behavior (continued)
QF-071	Bearer headers with valid tab separators failed parsing
QF-072	Admin token comparison used variable-time equality
internal/httpapi/server.go	
QF-073	Generated fallback quiz IDs shared the high-order clock prefix
QF-074	Integer query parameters accepted trailing junk
QF-075	Submission timestamps were sorted lexicographically
internal/middleware/middleware.go	
QF-076	Unbounded caller request IDs were echoed into responses
QF-077	Flushed responses could lose their correlation ID
QF-078	Log fields could overwrite the event identity and timestamp
QF-079	An unserializable log field silently erased the whole event
QF-080	Expired rate-limit clients accumulated indefinitely
QF-081	Retry-After rounded down and invited premature retry
QF-082	Integer formatting corrupted the minimum integer
internal/health/deep.go	
QF-083	Invalid error-rate measurements broke health JSON or reported healthy
QF-084	Impossible used-disk values were accepted
QF-085	Invalid load measurements broke health JSON or reported healthy
QF-086	Cancellation during the final health probe still reported healthy
QF-087	The standard load parser accepted nonfinite and negative readings
internal/chaos/state.go	
QF-088	A canceled zero-latency request still reached the handler
QF-089	An arbitrary Upgrade header bypassed injected faults
QF-090	Database-stall simulation omitted retake writes
QF-091	Informational HTTP statuses hid the final response status
QF-092	Flushing before writing recorded a different final status
web/src/components/quiz-preview-dialog.tsx	
QF-093	Loading previews had no status announcement
QF-094	Preview failures were not announced
QF-095	Preview options truncated the text teachers needed to check
QF-096	Preview prompts collapsed intentional line breaks
QF-097	Preview omitted question weights
QF-098	Preview omitted feedback teachers needed to verify
web/src/components/board/presence-panel.tsx	
QF-099	Participant initials split Unicode surrogate pairs
QF-100	Large participant lists extended beyond the viewport
QF-101	Whitespace participant names appeared blank
QF-102	The participant list contained invalid separator markup
QF-103	Truncated participant names could not be inspected
web/src/components/board/peer-cursors.tsx	
QF-104	White cursor labels had poor contrast on bright backgrounds
QF-105	Long cursor names could cover the canvas
web/src/hooks/use-mobile.ts	
QF-106	Mobile layout disagreed with the media query at viewport boundaries

ID	Corrected behavior (continued)
`web/src/hooks/use-online-status.ts`	
QF-107	Connectivity changes between render and effect were missed
QF-108	Restored pages kept stale connectivity state
`web/src/components/layout/app-shell.tsx`	
QF-109	Keyboard users had no way to skip repeated navigation
`web/src/components/layout/app-sidebar.tsx`	
QF-110	Account avatars split Unicode surrogate pairs
QF-111	Sign-out retained private query results in memory
QF-112	Active navigation lacked an accessible current-page indicator
QF-113	Mobile navigation stayed open after choosing a page
`web/src/routes/__root.tsx`	
QF-114	Initial client startup displayed a blank page
QF-115	Users with JavaScript disabled received no explanation
`web/src/styles.css`	
QF-116	Application animations ignored reduced-motion preferences
`web/src/lib/classrooms.ts`	
QF-117	Malformed cached invitation metadata crashed roster rendering
QF-118	Invalid cached class timestamps reached date rendering
QF-119	Invalid cached member timestamps broke roster ordering
QF-120	Classroom caches ignored changes made in other tabs
QF-121	Accepting a cached invitation retained the invitation's identity instead of the learner's
QF-122	An older classroom-list response could replace a newer list
QF-123	An older roster response could replace a newer roster
QF-124	An in-flight roster fetch undid a local membership mutation
QF-125	Saving class details overwrote concurrently refreshed members
QF-126	A late class update recreated a locally removed class
QF-127	A sign-in change between preflight and mutation sent the mutation as the new user
QF-128	Deleting a synced class missing from a limited catalog only deleted its local copy
QF-129	Leaving a synced class missing from a limited catalog only removed it locally
QF-130	An empty local join code matched an unsynced class
QF-131	Imperative selection returned IDs of missing classes
QF-132	Classroom hooks did not subscribe directly to identity changes
`web/src/components/classroom-dashboard-real.tsx`	
QF-133	Invitation links had duplicate slashes for origins ending in a slash
QF-134	Roster initials split Unicode surrogate pairs
QF-135	Repeated class write clicks could create overlapping operations
QF-136	The join dialog promised invitation URLs but submitted the entire URL as a code
QF-137	Joining a class removed unrelated query parameters and router history state
QF-138	Class dialogs could close while a write was still pending
QF-139	The create-class dialog did not submit with Enter
QF-140	The edit-class dialog did not submit with Enter
QF-141	The join-class dialog did not submit with Enter
QF-142	The invitation dialog did not submit with Enter
QF-143	Class-name fields allowed values beyond the server contract

ID	Corrected behavior (continued)
QF-144	Subject fields allowed values beyond the server contract
QF-145	Invitation names allowed values beyond the server contract
QF-146	Invitation emails allowed values beyond the server contract
QF-147	Teacher accounts attempted learner-only joins without an explanation
web/src/components/teacher/edit-exam-dialog.tsx	
QF-148	Catalog refreshes reset edits in an open settings dialog
QF-149	Repeated save clicks could overlap quiz-setting writes
QF-150	Invalid local datetimes silently cleared saved boundaries
QF-151	A late save response closed a different quiz's dialog
QF-152	Quiz settings could close through Escape while saving
QF-153	Quiz settings could not save through Enter
QF-154	A missing classroom option concealed the current assignment
web/src/components/confirm-action.tsx	
QF-155	Repeated confirmation events could run a destructive action twice
web/src/lib/teacher-authoring.ts	
QF-156	Duplicate CSV metadata headers silently selected the first value
QF-157	Extra CSV cells were silently discarded
QF-158	CSV scoring accepted hexadecimal and other unexpected number syntax
QF-159	A UTF-8 BOM prevented otherwise valid JSON imports
QF-160	Unbounded import text could freeze parsing and preview
web/src/components/teacher/question-import.tsx	
QF-161	Spellcheck altered the presentation of pasted import data
QF-162	Empty imports offered an action that could only fail
web/src/components/teacher/quiz-builder.tsx	
QF-163	Loading another draft left obsolete server work queued
QF-164	Undoing question removal could inject the question into a different draft
QF-165	Rapidly inserting the same bank question produced duplicate React keys
QF-166	Bank deduplication hid questions with distinct weights or feedback
web/src/routes/login.tsx	
QF-167	Redirect targets containing backslashes or control characters passed the local-path check
QF-168	Account-mode switches carried a visible password into the other mode
QF-169	Malformed login responses were treated as successful authentication
QF-170	Repeated authentication actions could overlap before React disabled them
QF-171	Successful password resets retained the revoked browser session
QF-172	Successful reset navigation left the used reset token in back history
QF-173	Authentication tabs could change while a request was in flight
web/src/routes/settings.tsx	
QF-174	Repeated profile-save events could overlap updates
QF-175	Repeated password-change events could overlap updates
QF-176	Password changes retained private query results after sign-out
internal/httpapi/server.go	
QF-177	CORS preflights did not explicitly allow bearer authorization
QF-178	Responses without an allowed Origin omitted the cache Vary key
QF-179	Cross-origin clients could not read Retry-After

ID	Corrected behavior (continued)
QF-180	Null and non-object JSON bodies were accepted by object endpoints
`internal/exams/exams.go`	
QF-181	Oversized or control-bearing request IDs reached persistence
QF-182	A missing service user could retrieve an exam
QF-183	Negative autosave versions were reported as storage outages
QF-184	Missing autosave request IDs were reported as storage outages
`web/src/routes/__root.tsx`	
QF-185	Reset-link query tokens could be sent in resource referrers
`internal/continuum/router.go`	
QF-186	An unconfigured router panicked instead of returning unavailable
QF-187	Routing-state reads were outside the request timeout
QF-188	Backend redirects bypassed the selected routing target
QF-189	Requests with a nil body panicked in the router
QF-190	Base64 request decoding allocated the entire oversized payload before checking its limit
`internal/continuum/probe.go`	
QF-191	Health probes followed redirects to unrelated destinations
QF-192	Health probes accepted trailing JSON documents or garbage
`internal/continuum/local.go`	
QF-193	Timed-out local-demo shutdown returned before closing its listeners
`internal/emaildelivery/smtp.go`	
QF-194	Display-name sender addresses were sent as invalid SMTP envelopes
QF-195	Unicode invitation subjects were sent without MIME encoding
QF-196	Unicode email bodies relied on unadvertised eight-bit transport support
`web/src/routes/teacher.tsx`	
QF-197	A failed analytics catalog had no retry action
QF-198	A failed selected quiz trapped analytics without a selector or retry
QF-199	An assessment with no submissions displayed a misleading zero-percent average
QF-200	Assessment progress bars lacked accessible names